University of Nebraska-Lincoln
School of Computing

CSCE 474/874: Introduction to Data Mining
Spring 2026

Homework 4 March 5, 2026

Assignment

Implement the *k*-means algorithm to perform clustering and compare your results with the results from Weka.

- Assume that all the attributes are continuous variables.
- Your program must allow the number of clusters (*k*) to be specified as input.
- Your program must allow the epsilon (change in the sum of the distances from the cluster centers) to be specified as input.
- Your program must allow the number of iterations to be specified as input.

Your program should stop if either the number of iterations is reached or if the change in the total sum of the squares of the distances (SSD) falls below epsilon. After implementing the algorithm, perform the following.

1. Plot the runtime of the algorithm as a function of number of clusters, number of dimensions and size of the dataset (number of transactions).
2. Plot the goodness of clustering as a function of the number of clusters and determine the optimal number of clusters.
3. Compare the performance of your algorithm with that of Weka and summarize your results.

You will work with your project team members to complete this assignment.

Use the dataset you will be working on for the project. If the data is not suitable, you may use one from the Weka dataset.

All code must be written from scratch by the members of your team. You may NOT use any code from ANY OTHER source, including other students, AI tools, and the Internet. You may NOT use predefined functions provided in any package/library.

Due Date

The assignment is due on March 24 is worth 70 points. **However, if you complete the assignment by March 15, you will get 10 bonus points.**

Handin

Hand in a report along with the listing of your program, the output generated from the run of the test file on Canvas. Make sure that you have uploaded a signed copy of the Contributions form. Prepare and submit two files as follows:

- Your report named as “Lastname1_Lastname2_Lastname3_Lastname_4.pdf” in pdf format. The signed contributions form should be used as the cover page of your report.
- A zip file named “Lastname1_Lastname2_Lastname3_Lastname_4.zip” that includes everything else (your program, the output generated from the run of the test file, etc.). You must include a README file that describes the usage of your program.

Grading Guidelines

Implement the *k*-means algorithm to perform clustering in a dataset. (30 points)

- Implement the algorithm and provide specific and complete instructions on how to test your code in the README file. (25 points)
- In the report, you should write a paragraph about your program design, list of functions in your code and their specific roles (5 points)

Plot the runtime of the algorithm as a function of number of clusters, number of dimensions and size of the dataset (number of transactions). (15 points)

- In the report, you should write a paragraph to summarize the graphs and elaborate on it.

Plot the goodness of clustering as a function of the number of clusters and determine the optimal number of clusters. (15 points)

- In the report, you should write a paragraph to summarize the graphs and elaborate on it.

Compare the performance of your algorithm with that of Weka and summarize your results. (10 points)

- Summarize the differences (if there is any) and elaborate on it (why/how).